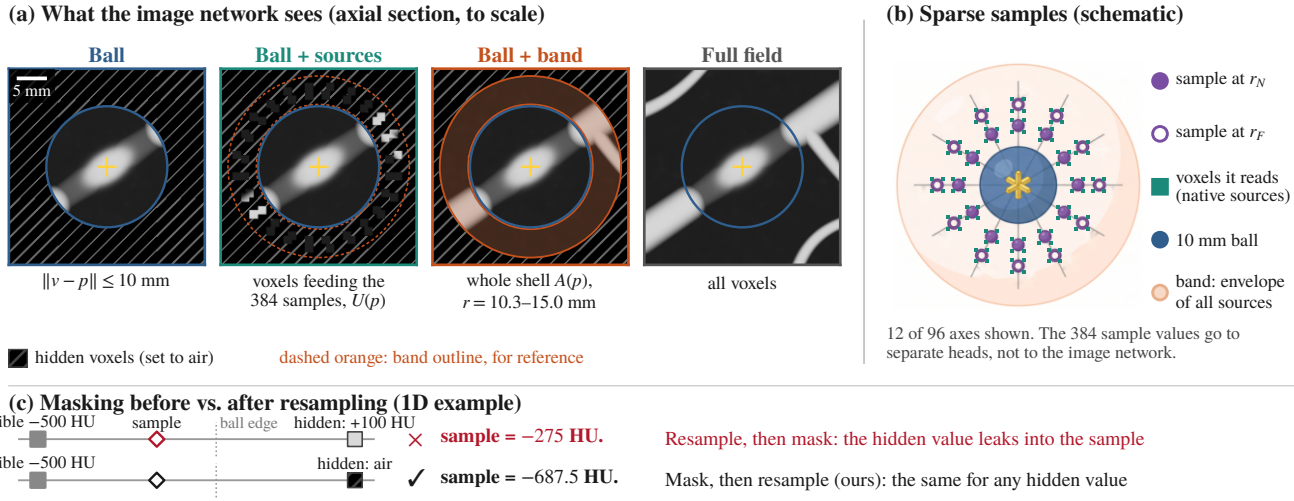

(a) What the image network sees (axial section, to scale)
Ball
Ball + sources
Ball + band
Full field
5 mm
$\|v - p\| \le 10$ mm
voxels feeding the 384 samples, $U(p)$
whole shell $A(p)$, $r = 10.3$–15.0 mm
all voxels
hidden voxels (set to air)
dashed orange: band outline, for reference
(b) Sparse samples (schematic)
sample at r_N
sample at r_F
voxels it reads (native sources)
10 mm ball
band: envelope of all sources
12 of 96 axes shown. The 384 sample values go to separate heads, not to the image network.
(c) Masking before vs. after resampling (1D example)
visible −500 HU
sample
ball edge
hidden: +100 HU
sample = −275 HU.
visible −500 HU
hidden: air
sample = −687.5 HU.
Resample, then mask: the hidden value leaks into the sample
Mask, then resample (ours): the same for any hidden value